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Bronchoalveolar lavage in drug-induced lung disease

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Since its first application in the 1970s [1] as a valuable research tool to study local immune and inflammatory mechanisms [2,3], bronchoalveolar lavage (BAL) has gained widespread acceptance as a powerful investigative tool of pulmonary medicine and has become a standard diagnostic procedure in patients who have interstitial lung disease [4–7]. BAL is used to collect cells, inhaled particles, infectious organisms, and solutes from the lower respiratory tract, and, in particular, from the alveolar spaces of the lung [8]. To achieve this, a sufficient volume of lavage fluid must be instilled to ensure a sufficient aspirate. In adults, a minimum of 100 mL instillation should be made [9].

Diagnostic BAL serves as a "window to the lung." BAL is considered to provide complimentary information to histopathology from biopsies, but has several advantages over biopsy procedures. It is safe and associated with little to no morbidity. Thus, it can be used repeatedly to investigate serial changes. In addition, BAL collects samples from a much larger area of the lungs than can be obtained by the small tissue fragments of transbronchial biopsy or even by surgical biopsy specimen; therefore, it gives a more representative view of inflammatory and immunologic changes. It is believed that alterations in lavage fluid and cells reflect pathologic changes in the corresponding parenchymal constituents. Several studies showed a good correlation between the type and number of inflammatory cells obtained by BAL and those observed in histologic sections of lung biopsy specimen or derived from mechanically-disaggregated lung tissue in several interstitial lung disorders, such as idiopathic pulmonary fibrosis, sarcoidosis, and hypersensitivity pneumonitis [10–14].

This article reviews some technical aspects of BAL, gives an overview on the use of BAL in the differential diagnosis of interstitial lung disease, and concludes with a detailed description of BAL findings in drug-induced lung disease.

Technical aspects

Guidelines and recommendations for a standardized approach to the performance of BAL and the laboratory processing and the analysis of recovered fluid have been published [4,9,15,16]. When the guidelines are followed, the results of lavage are valid for practical diagnostic purposes.

Bronchoscopic procedure

BAL usually is performed during fiberoptic bronchoscopy with topical anesthesia following general inspection of the tracheobronchial tree. The fiberoptic bronchoscope is introduced gently until impacted, or "wedged," in a segmental or subsegmental bronchus. In diffuse lung disease, the middle or lingular lobe is recommended as a standard site of BAL. Localized disease requires lavage of the radiographically-involved area.

Commonly, sterile, unbuffered isotonic saline (0.9% NaCl solution) is used as the instillate. The fluid is instilled with syringes through the biopsy channel of the bronchoscope, using multiple aliquots of 20 mL to 60 mL (four to five aliquots are the most common), to a total volume of 100 mL to 300 mL. Smaller instilled volumes (less than 100 mL) increase the likelihood of contamination of lavage

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fluid by the bronchial spaces, including inflammatory cells from this compartment, which may influence the differential cell counts [17]. After each instillation, the aliquot is recovered immediately by gentle hand suction into the syringe or gentle wall suction into a fluid trap. Usually, 40% to 70% of the instilled volume is recovered.

BAL generally is well tolerated. Side effects are similar to those of routine fiberoptic bronchoscopy under local anesthesia. There is practically no mortality and the complication rate ranges between 0% and 2.3%, compared with 7% with transbronchial biopsy and 13% with surgical lung biopsy [18,19]. There are no absolute contraindications for performance of BAL beyond those noted for bronchoscopy.

Laboratory processing

The total recovered fluid should be transferred to the laboratory as quickly as possible because the cells are not well-preserved in the saline solution [20]. The total number of cells is counted in a hemocytometer, either in a sample of the pooled native fluid or in a resuspension of the cells after the first centrifugation. Cell viability is assessed by trypan blue exclusion and should range from 80% to 95%.

For the enumeration of cell differentials, at least 600 cells are counted after applying May-Grünwald-Giemsa stain. A high percentage of epithelial cells (> 5%) is indicative of contamination of the alveolar samples by bronchial cells. Such BAL probes may not be representative for the diagnosis of diffuse parenchymal lung disease. Several soluble components have been measured in BAL fluid. None of them has proven to be useful in clinical settings [21].

Normal findings in healthy adults who do not have lung disease

The BAL fluid that is obtained from healthy, nonsmoking adults who do not have lung disease contains only small percentages of lymphocytes, neutrophils, and other inflammatory cells [4,5,15]. The alveolar macrophages are the predominant cell population.

Cigarette smoking is a strong confounding factor with significant effects on BAL samples [15,22,23]. The alveolar macrophages from smokers show a characteristic appearance; many are much larger than those in nonsmokers and contain cytoplasmic inclusion bodies (smoker's inclusion bodies). On gross examination, the recovered BAL fluid has a light to dark brown and turbid appearance that is caused by the tar-laden macrophages. The total cell yield is threefold to fivefold higher in smokers because of a threefold to fivefold increase in the number of macrophages; this results in a relative decrease in the percentage of lymphocytes.

The normal values of differential cytology following BAL that have been proposed in the literature are variable. In most of the published studies, only small numbers of normal patients/persons (mostly volunteers) are available for comparison [15]. For practical reasons, the following percentages can be expected as normal in nonsmokers [9]:

Macrophages greater than 80%
Lymphocytes up to 15%
Neutrophils up to 3%
Eosinophils up to 0.5%
Mast cells up to 0.5%

Bronchoalveaolar lavage as a clinical diagnostic tool

BAL is indicated in every patient who has unclear interstitial lung disease or unclear pulmonary shadowing, no matter what cause is suspected. The underlying disorders may be of infectious, noninfectious, immunologic, or malignant origin. BAL also may be indicated in patients who have normal chest radiograms when clinical and lung function tests are abnormal and point toward a diffuse lung disease or in patients who have unexplained pulmonary symptoms and in whom a normal BAL finding may allow the exclusion of significant, active interstitial lung disease.

Interpretation of bronchoalveolar lavage cytology

Changes in the profile of the cell differentials of BAL fluid have been described in various interstitial lung diseases. Clearly, an abnormal BAL cell differential count does not allow a specific diagnosis, as it does with conventional differential blood counts; however, BAL studies should not be limited to counting the cell differentials. At least as important as examining cell differentials is to note the morphologic appearances of cells and particles. Examples are the different morphology in hypersensitivity pneumonitis (foamy macrophages, heterogeneous macrophage size, presence of plasma cells) versus that of sarcoidosis (more monomorphous appearance of macrophages, less activated lymphocytes), the presence of malignant cells, the characteristic features of alveolar proteinosis, or the detection of dust particles, such as asbestos bodies in occupational exposure conditions [24].

Furthermore, BAL cell differentials should not be used as an isolated finding for making a diagnosis but always should be interpreted in the context of disease history and clinical, laboratory, and radiologic findings.

Bronchoalveolar lavage in differential diagnosis of interstitial lung diseases

BAL has improved the diagnostic work-up; some investigators use BAL more frequently than transbronchial lung biopsy. In two recently-published international statements on the major interstitial lung diseases, BAL was considered to be helpful in strengthening the diagnosis in patients who had sarcoidosis in the absence of biopsy [25]; BAL or transbronchial biopsy were considered requirements for the exclusion of other diseases in a patient who had idiopathic pulmonary fibrosis (IPF) who did not undergo surgical biopsy (one of the four major criteria for making a clinical diagnosis of the disease) [26].

BAL findings may, on occasion, be extremely specific, so that they can directly confirm a particular diagnosis and replace lung biopsy (Table 1). Many of these disorders are included in the group of the alveolar filling syndromes. Obviously, the abnormal material that accumulates in the alveolar spaces in these syndromes can be washed out easily by lavage and is so characteristic that a specific BAL diagnosis is possible.

In other selected lung diseases, BAL findings are not diagnostic but may help narrow the differential diagnosis in the appropriate clinical setting if the BAL results fit with the clinician's suspicion and with the imaging results. The list of diseases that have a lymphocytic, neutrophilic, eosinophilic, or mixed cellular pattern is long (Table 2). Diseases with an

Table 1
Diagnostic bronchoalveolar lavage findings

BAL finding	Diagnosis
Pneumocystis carinii, fungi, cytomegalovirus transformed cells	Opportunistic infections
Milky effluent, periodic acid-schiff positive noncellular corpuscles, amorphous debris, foamy macrophages	Alveolar proteinosis
Hemosiderin-laden macrophages, intracytoplasmic fragments of red blood cells in macrophages, free red blood cells	Alveolar hemorrhage syndrome
Malignant cells of solid tumors, lymphoma, leukemia	Malignant infiltrates
Dust particles in macrophages, quantifying asbestos bodies	Dust exposure
Eosinophils greater than 25%	Eosinophilic lung disease
Positive lymphocyte transformation test to beryllium	Chronic beryllium disease
CD1 positive Langerhans cells increased	Langerhans cell histiocytosis
Atypical hyperplastic type II pneumocytes	Diffuse alveolar damage, drug toxicity

Table 2
Bronchoalveolar lavage cellular patterns as an adjunct to diagnosis (data from ref. 24)

Cellular pattern	Diagnosis
Lymphocytic	Hypersensitivity pneumonitis
	Berylliosis
	Sarcoidosis
	Tuberculosis
	Connective-tissue disorders
	Drug-induced pneumonitis
	Malignant infiltrates
	Silicosis
	Early asbestosis
	Crohn's disease
	Primary biliary cirrhosis
	HIV infection
	Viral pneumonia
Neutrophilic (± eosinophilic)	Idiopathic pulmonary fibrosis
	Desquamative interstitial pneumonia
	Acute interstitial pneumonia
	Acute respiratory distress syndrome
	Bacterial pneumonia
	Connective-tissue disorders
	Asbestosis
	Wegener's granulomatosis
	Diffuse panbronchiolitis
	Transplant bronchiolitis obliterans
	Idiopathic bronchiolitis obliterans
	Drug-induced reaction
Eosinophilic	Eosinophilic pneumonia
	Churg-Strauss syndrome
	Hypereosinophilic syndrome
	Allergic bronchopulmonary aspergillosis
	Idiopathic pulmonary fibrosis
	Drug-induced reaction
Mixed cellularity	Bronchiolitis obliterans organizing pneumonia (BOOP)
	Connective-tissue disorders
	Nonspecific interstitial pneumonia
	Drug-induced reaction
	Inorganic dust disease

Data from Costabel U. Atlas of bronchoalveolar lavage. London: Chapman and Hall; 1998.

Table 3
CD4/CD8 ratio in diseases with lymphocytic alveolitis

CD4/CD8 increased	CD4/CD8 normal	CD4/CD8 decreased
Sarcoidosis	Tuberculosis	Hypersensitivity pneumonitis
Beryllium disease	Lymphangitic carcinomatosis	Drug-induced pneumonitis
Asbestos-induced alveolitis		BOOP
Alveolar proteinosis		Silicosis
Crohn's disease		HIV infection
Connective-tissue disorders		

increase in lymphocytes can be differentiated further into those with an elevated, normal, or decreased CD4:CD8 ratio (Table 3). In these settings, combining BAL with high resolution computed tomography (HRCT) has increased the diagnostic power. For example, if the CT scan shows a patchy ground-glass pattern, BAL may be able to reveal that this patient suffers from extrinsic allergic alveolitis (high lymphocyte count), a smoking-related respiratory bronchiolitis/interstitial lung disease (high smoker macrophage count and normal cell differential), or alveolar hemorrhage (high count of hemosiderin-laden macrophages). Sometimes, a normal lavage may be useful to exclude some disorders with high probability (eg, extrinsic allergic alveolitis, eosinophilic pneumonia, alveolar hemorrhage) and to focus attention in other directions.

Assessment of disease activity

It has not been established whether BAL is clinically useful for the assessment of the activity of disease processes and to provide prognostic information. It also is not clear that BAL is better than other parameters to guide therapy. In this regard, serial BAL to monitor the course of disease cannot be recommended routinely.

Bronchoalveaolar lavage in drug-induced pneumonitis

Many drugs can induce a toxic or immune-mediated interstitial lung reaction. In addition to toxic changes of the type II pneumocytes, any type of alveolitis (lymphocytic, neutrophilic, eosinophilic, or mixed) and diffuse alveolar hemorrhage may be present in BAL (Box 1) [4,24,27–29]. The most frequent finding is a lymphocytic alveolitis with the predominance of $CD8^+$ cells, just as in extrinsic allergic alveolitis. In methotrexate-induced pneumonitis, the $CD4^+$ cells may be increased preferentially [30,31]; however another study reported the opposite [32]. Table 4 lists, in detail, the changes in BAL cytology that has been reported for individual drug.

Amiodarone causes characteristic changes in the alveolar macrophage population, which shows foamy intracytoplasmic alterations and corresponds to a form of phospholipidosis [27,28]. The drug induces an increase in the phospholipid content of all body cells, and, especially, phagocytic cells. Electron microscopy shows that the vacuoles in the cytoplasm correspond to large numbers of surfactant-like lamellar bodies, which is consistent with the phospholipid nature of these inclusions. This feature also is seen in patients who are treated with amiodarone but are free of clinical lung involvement. If foamy macrophages are not present in BAL, an amiodarone-induced pneumonitis likely can be excluded. Conversely, a normal BAL differential does not exclude amiodarone-induced lung disease [37–39].

BAL findings are not, however, specific for any drug-induced lung disease; the definitive diagnosis needs additional clinical investigations. BAL is useful, however, to exclude other pulmonary problems that commonly are associated with chemotherapeutic agents.

Diagnostic features of drug-induced lung disease

The diagnosis of drug-induced lung disease usually is based on several criteria [103], including:

- A history of drug exposure
- Clinical, imaging, and histopathologic patterns that are consistent with earlier observations with the same drug
- Exclusion of other lung disease
- Improvement following discontinuation of the suspected drug
- Recurrence of symptoms on rechallenge (this may be done in certain circumstances, but usually is considered unethical in drugs that cause irreversible pulmonary changes) [103]

BAL may contribute to several of these criteria. The characteristic changes of amiodarone (macro-

Box 1. Overview on BAL in drug-induced interstitial lung disease. For a detailed listing see Table 4

Lymphocytosis

Methotrexate
Azathioprine
Cyclophosphamide
Bleomycin
Busulfan
Vincristine
Nitrofurantoin
Minocycline
Gold
Sulfasalazine
Amiodarone
Acebutolol
Atenolol
Celiprolol
Propanolol
Flecainide
Diphenylhydantoin
Nilutamide

Neutrophilia

Bleomycin
Busulfan
Minocycline
Amiodarone

Eosinophilia

Bleomycin
Nitrofurantoin
Cotrimazole
Penicillin
Sulfasalazine
Ampicillin
Tetracycline
Maloprim
Minocycline
L-tryptophan

Haemorrhage

D-penicillamine
Amphotericin B
Cytotoxic drugs

Cytotoxic reaction

Bleomycin
Methotrexate
Nitrosureas
Busulfan
Cyclophosphamide

phages with foamy, vacuolated cytoplasm) are evidence of exposure to the drug. If these changes are missing, the interstitial lung disease probably is not associated with amiodarone.

Bronchoalveolar lavage to support a certain type of drug-induced lung disease

BAL may support a certain clinical-pathologic pattern of lung involvement:

Cellular pneumonitis, also called "hypersensitivity pneumonitis," is the most common drug-induced interstitial lung disease. BAL shows the characteristic pattern with an isolated and marked increase in lymphocytes, usually greater than 50%. A moderate increase in neutrophils and other cells also may be present. Usually, the CD4:CD8 ratio is low [33,46,82,87,93].

Pulmonary infiltrates with eosinophilia or eosinophilic pneumonia is a reaction that is associated with many drugs. BAL shows significant eosinophilia [33]. The absence of BAL eosinophilia is rather exclusive of pulmonary eosinophilic infiltrates.

Organizing pneumonia is another pattern that is associated with several drugs. BAL usually shows a mixed pattern with increases in lymphocytes, neutrophils, eosinophils, and mast cells [104]. Foamy alveolar macrophages and a small number of plasma cells also can be seen. The CD4:CD8 ratio is low.

A cytotoxic reaction, associated with cytostatic chemotherapy, may show histologic changes, such as hyaline membranes, atypia and hyperplasia of type II pneumocytes, and diffuse fibroblast proliferation (ie, a similar pattern to diffuse alveolar damage). BAL may show clusters of atypical type II pneumocytes and a marked increase in neutrophils; this is seen commonly in patients who have acute respiratory distress syndrome or acute interstitial pneumonia [105].

Diffuse alveolar hemorrhage is associated with a few drugs. BAL is the ideal technique to demonstrate the hemosiderin-laden alveolar macrophages, in addition to the presence of free red blood cells.

Bronchoalveolar lavage to exclude other diseases

The major role of BAL is to exclude infections; this is essential in patients who have acute febrile

Table 4
Type of changes in bronchoalveolar lavage cytology reported for individual drugs

Drug name	Ly	Neu	Eos	Mixed	Hem	Cytotox	CD4/CD8	References
Acebutolol	X							[33]
Acetaminophen	X						low	[34]
Acetylsalicylic acid (aspirin)			X					[35,36]
Amiodarone[a]	X	X	X	X			low	[28,37–39]
Amitriptyline			X					[40]
Amphotericin B					X			[33]
Ampicillin	X		X				high	[33,41]
Atenolol	X			X				[24]
Azathioprine	X				X		normal	[42,43]
Barbiturates			X					[44]
BCG therapy (intravesical)	X						high	[33,45]
Bleomycin	X	X	X			X		[33,46]
Bromocriptine	X	X						[47]
Bucillamine	X		X				low	[48,49]
Busulfan	X			X		X		[50]
Carbamazepine	X	X				X	low	[51–53]
Celiprolol	X							[54]
Chlorambucil	X						low	[55]
Chloroquine			X				low	[56]
Co-trimoxazole				X			low	[57]
Cyclophosphamide	X					X		[58]
Cyclothiazide	X							[59]
Cyproterone acetate				X			low	[60]
Disopyramide	X							[61]
Etoposide	X							[62]
Flecainide	X			X			low	[33,63]
Gold (Auranofin)	X			X			low, high	[64,65]
Heroin			X					[66]
Interferon-α	X			X			low	[67]
Interferon-β				X			low	[68]
Ifenprodil			X					[69]
Indomethacin			X					[70]
Isoniazid				X				[71]
Levofloxacin			X				low	[72]
Loxoprofen			X					[73]
Maloprim (pyrimethamine and dapsone)			X					[74]
Melphalan	X							[75]
Methotrexate	X		X	X		X	high	[30,31,76]
Minocycline	X	X	X				low	[77,78]
Mitomycin C	X						normal	[79]
Nalidixic acid			X					[80]
Nilutamide	X	X					low	[81,82]
Nitrofurantoin	X		X				high	[83]
Nitrosureas (Carmustine)	X					X	low	[84]
Oxaliplatin			X			X		[85,86]
Paclitaxel	X		X				low	[87]
Penicillamine				X	X			[33]
Penicillin			X					[33]
Piperacillin	X						low	[88]
Phenytoin	X							[89,90]
Pranoprofen			X					[91]
Procainamide		X						[92]
Propranolol	X						low	[93]
Rifampicin	X							[94]

(continued on next page)

Table 4 (*continued*)

Drug name	Ly	Neu	Eos	Mixed	Hem	Cytotox	CD4/CD8	References
Serrapeptase	X		X				low	[95,96]
Sirolimus	X				X		high	[97]
Sulfasalazine	X		X					[33,98]
Tetracyclin			X					[33]
Tolfenamic acid	X		X				high	[99]
Tosufloxacin tosilate	X	X	X					[100]
Trazodone			X					[101]
Tryptophan			X					[24,102]
Vincristine	X							[24]

Abbreviations: BCG, bacillus Calmette-Guerin; cytotox, cytotoxic reaction; eos, eosinophilia; hem, alveolar hemorrhage; ly, lymphocytosis; mixed, mixed cellularity; neu, neutrophilia.

[a] Foamy alveolar macrophages.

interstitial lung disease and in those who receive immunosuppressive therapy. It is important to exclude opportunistic infections, because methotrexate, corticosteroids, immunosuppressives, and infliximab have been associated with the development of bacterial, fungal, parasitic (including *Pneumocystis carinii*), and viral infections. BAL has a high sensitivity and specificity (>95%) to diagnose *P carinii* pneumonia. Globally, the sensitivity of BAL in the diagnosis of bacterial pneumonia is 60% to 90%; it is 70% to 80% in mycobacterial, fungal, and viral infections.

Many cardiovascular drugs have been reported to induce interstitial lung disease. The differential diagnosis of left heart failure with pulmonary congestion is a frequent consideration. The diagnosis of pulmonary congestion due to left heart failure can be supported by the BAL finding of diffuse alveolar hemorrhage. It is not known that cardiovascular drugs can induce diffuse alveolar hemorrhage (except for anticoagulants). Conversely, if the BAL cytology is normal it largely can exclude a drug-induced immunologic reaction (eg, hypersensitivity pneumonitis, organizing pneumonia, eosinophilic lung disease).

Bronchoalveolar lavage in the diagnosis of underlying disorders

Pulmonary involvement by a malignancy or a systemic disease, such as connective tissue disorders and inflammatory bowel disease, is important. BAL can contribute to the diagnosis of metastatic adenocarcinoma with a sensitivity of 80% and lymphoma with a sensitivity of 50% [106]. Interstitial lung disease that is associated with connective tissue disease or inflammatory bowel disease has nonspecific BAL findings; any type of BAL pattern (lymphocytic, neutrophilic, eosinophilic, and mixed) may be present so a differentiation between manifestations of systemic immunologic disorders and immunologic drug-induced disorders is not possible by BAL.

Summary

Given the varying histologic reactions and different mechanisms of action, it is not surprising that no uniform constellation of BAL changes is seen in drug-induced lung disease. BAL findings are not specific for any drug-induced lung disease and the definitive diagnosis cannot rely solely on the BAL findings. BAL findings can, however, contribute to the expected clinicopathologic pattern of a given drug-induced lung disease. BAL also is helpful in the differential diagnosis, primarily in the exclusion of an infective cause and of involvement of the lungs by the underlying disease (eg, metastatic cancer or malignant lymphoma).

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